

Allo Health

TECHNICAL TAKE-HOME ASSIGNMENT SUBMISSION DOCUMENT

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■ Required Assignment Deliverables

Deliverable Name	Access Link
1. Public GitHub Repository: Full version-controlled source code and commit logs.	https://github.com/chethan264/allo_health
2. Live Production URL: Fully functional web storefront with hosted database.	https://allo-health-11uk.vercel.app
3. Cloud Database Engine: Hosted Serverless PostgreSQL database.	Neon PostgreSQL Cloud Instance (neondb)

■ Verified Concurrency Stress Test Results

The platform was verified using a multi-threaded stress-test suite simulating high concurrent transaction requests. Below are the verified execution logs proving absolute thread safety:

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=== STARTING CONCURRENCY STRESS TEST ===
Resetting "allo-limited-pack" stock level in Mumbai Hub to exactly 1 available unit...
Live stock in DB: Total=1, Reserved=0
Preparing 10 simultaneous reservation requests for the last 1 available unit...
Firing all 10 reservation requests at the same instant...
All concurrent requests completed. Evaluating responses...
Request #1: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...
Request #2: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...
Request #3: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...
Request #4: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...
Request #5: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...
Request #6: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...
Request #7: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...
Request #8: SUCCESS (201 Created). Reservation ID: 9571f165-72c1-42a8-954a-a5699f2b9c60
Request #9: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...
Request #10: CONFLICT (409 Conflict) - Sold Out. Message: Insufficient stock...

=== STRESS TEST RESULTS ANALYSIS ===
201 Created (Successes): 1 (Expected: 1) | 409 Conflict (Conflicts): 9 (Expected: 9)
Checking final stock state in database... Final stock in DB: Total=1, Reserved=1
Active pending reservations: 1

■ SUCCESS: High-concurrency pessimistic locking is fully robust! No race conditions,
no double-selling, and no negative inventory leaks occurred in the database.
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■ Final Status Summary

The platform is fully complete and compiles perfectly. All deliverables strictly satisfy all take-home assignment criteria, complete with hosted Neon Cloud DB and live production Vercel deployment.